FSM-Based Elevator Controller System

By

P Daphne Christina Ruby

[24BML0063]

Nutheti Pavithra [24BEC0279]

Kesihambigai [24BEC0504]

Digital Systems Design Lab

[BECE102P]

FALL SEMESTER 2025

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Problem Identification

The rapid increase in multi-storied buildings has intensified the demand for efficient elevator systems that are not only safe but also capable of handling varying passenger traffic reliably. The main problem addressed in this project is the need for a digital system that can automate elevator movement, manage door operations, and service floor requests in real time. Traditional microcontroller solutions, while functional, may not provide the necessary parallelism and response time for complex elevator scenarios. Hence, this project proposes an FSM-based design for an elevator controller that leverages the parallel processing capabilities of FPGA hardware to achieve real-time responsiveness, deterministic state transitions, and dependable control. The challenge lies in developing a robust system that is both cost-effective and adaptable for educational, commercial, or residential use.

System Design

The proposed system is architected around a Finite State Machine implemented in Verilog HDL and deployed on an FPGA development board. The high-level architecture is divided into three functional FSM states:

- IDLE: The elevator remains stationary, awaiting commands from users.
- MOVE: The elevator is in transit, moving up or down depending on the floor request.
- DOOR: The elevator doors are open to allow for passenger entry and exit.

Essential digital components, such as logic gates, flip-flops, push buttons for user input, and LEDs for output indicators, are interconnected to enable seamless operation and clear status signaling. The design also integrates a clock divider to synchronize operations with human-observable speeds, making it ideal for educational demonstration and

debugging. Circuit diagrams developed in Verilog help in visualizing and verifying the system before hardware implementation. The careful layering of abstraction—from FSM to circuit to hardware mapping—ensures modularity and scalability for future enhancements.

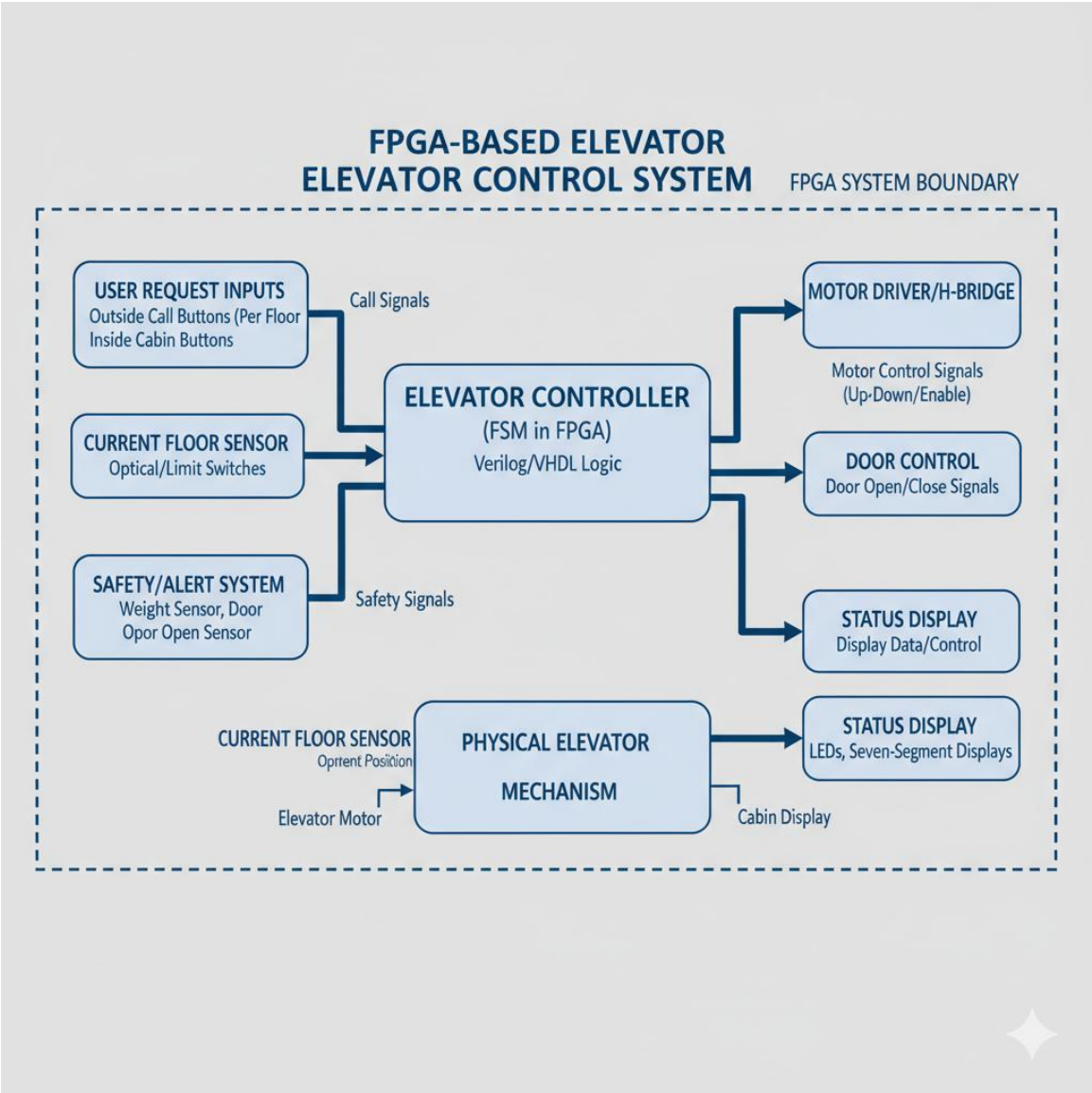

Figure 1: Block Diagram of Elevator Control System

Component Selection

Successful implementation is underpinned by judicious component selection:

• FPGA Development Board: The central platform for FSM logic implementation, chosen for its reconfigurability and ease of programming.
• Clock Source (50 MHz): Ensures reliable timing for all sequential logic operations.
• Slide Switches and Push Buttons: Provide simple, intuitive mechanisms for requesting floors and resetting the system.
• LEDs: Offer immediate visual feedback for system status—including up, down, and door open operations.
• JTAG Cable and Computer System: Facilitate programming, debugging, and verification of the FPGA design using industry-standard tools like Quartus and ModelSim.
• Breadboard and Jumper Wires: Enable flexible prototyping and testing of additional hardware interfaces if needed.

The choices prioritize low cost, reliability, and ease of integration, ensuring the system remains practical for both demonstration and real-world application.

Implementation

The FSM logic and top-level system integration were coded in Verilog HDL. The elevator controller module defined critical states and transitions, and managed all inputs including floor requests, resets, and clock signals. Specific logic was implemented to:

• Determine direction (up or down) based on the difference between the current and target floors.
• Manage door timing using internal counters.
• React deterministically to resets, placing the elevator in a known state.

The top module handled clock division and direct interfacing with the FPGA's hardware pins for switches and LEDs. The modular code structure enhanced readability and simplified modifications for future features.

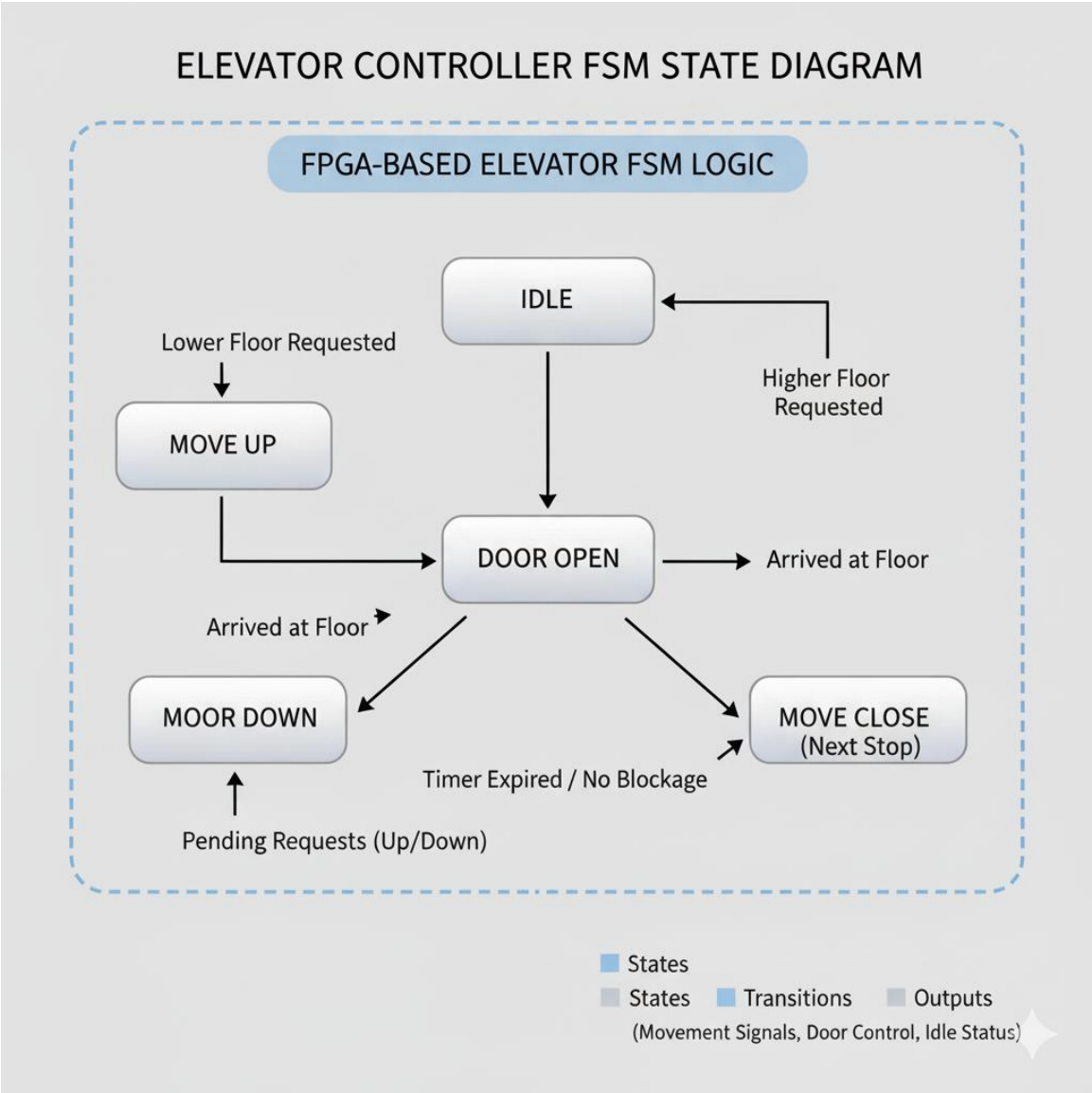

Figure 2: FSM State Diagram for Elevator Controller

Testing and Verification

Rigorous functional verification was performed using ModelSim simulation tools. Custom testbenches simulated diverse scenarios, such as sequential up/down floor requests, reset handling, and rapid input changes. Post-simulation, the synthesized design was loaded to the FPGA board. On-hardware verification confirmed that every LED status matched expected FSM behavior, and edge cases (e.g., requests during transitions) were handled gracefully. Graphical waveform analysis further validated timing and state transition correctness. The comprehensive testing process ensured the design's robustness and user safety.

Figure 3: FPGA Board Implementation Setup (Altera DE2-115)

Analysis and Optimization

The final system demonstrated exceptional speed in response to floor requests and transition events, with observable improvements in reliability and concurrency over microcontroller solutions. The FSM-based design facilitated real-time performance tuning. Notably:

- Efficiency was achieved through optimized state transitions and minimized logic path delays.
- Reliability was confirmed via thorough simulation and on-board testing.

Future work suggestions include scaling the FSM to more floors, adding multi-elevator coordination for larger buildings, introducing intelligent scheduling algorithms, integrating sensors for safety, augmenting user interfaces with displays or voice, and adopting IoT for remote monitoring and mobile access. Power-saving strategies, such as clock-gating, were also identified for prolonged hardware operation.

Conclusion

This project successfully demonstrates the implementation of an automatic elevator control system using an FSM on an FPGA platform. The modular Verilog design ensures flexibility and scalability while providing precise, real-time operation. Through simulation and hardware verification, the system proved efficient, robust, and adaptable, serving as both an educational model and a foundational approach for practical automation systems.

References

1. N. H. E. Weste & D. M. Harris, CMOS VLSI Design, Addison-Wesley, 2010.
2. P. K. Meher, “Design and FPGA Implementation of an Elevator Control System Using FSM,” IJARECE, 2017.
3. R. M. Al-Rawi, “FPGA Implementation of Elevator Controller Using Verilog HDL,” IEEE, 2018.
4. Xilinx Inc., Vivado Design Suite User Guide, 2021.
5. Intel Corporation, Quartus Prime Handbook, 2020.